

|           |                                                                                                                                                                                                                                                                                                                                                                                                                   |   |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1(a)(i)   | DOPA / dopaquinone / melanin ;                                                                                                                                                                                                                                                                                                                                                                                    | 1 |
| 1(a)(ii)  | cyclic AMP / cAMP ; <b>A</b> calcium ions                                                                                                                                                                                                                                                                                                                                                                         | 1 |
| 1(a)(iii) | <p>any <b>two</b> from:</p> <p>1 albino / albinism ;</p> <p>2 no pigment or melanin in skin / pale skin<br/><b>or</b><br/>white hair ; <b>Ignore</b> fair, skin / hair</p> <p>3 red / pink, eyes / irises ; <b>A</b> blue</p> <p>4 poor vision / described<br/><b>or</b> eyes sensitive to light<br/><b>or</b> jerky eye movements ; <b>Ignore</b> blindness</p>                                                  | 2 |
| 1(b)      | <p>symbols ; e.g. <b>B</b> <u>allele</u> for black fur<br/><b>b</b> <u>allele</u> for (light) brown fur</p> <p>parental genotypes <b>Bb</b> x <b>Bb</b></p> <p>gametes <b>B</b> <b>b</b> <b>B</b> <b>b</b> ] ;</p> <p>offspring genotypes <b>BB</b> <b>Bb</b> (<b>Bb</b>) <b>bb</b> ;</p> <p>offspring phenotypes black black (black) (light) brown ;</p> <p><i>accept gametes mark from a Punnett square</i></p> | 4 |